Data Distribution

It is the process of analyzing data in terms of data behaviour to uncover data patterns and ~~podcast~~ forecast future outcome. It is also helpful in data characterisation.

It is of Different Types:-

1. Normal Data Distribution (continous prob dis.
2. Binomial " "
3. Poission " "
4. Uniform & Exponential " "

Q. Mean, Median, Mode, Variance calulation, on a numerical dataset

Data Visualization

1. Bar Chart
2. Line chart
3. Scaratted Plots
4. Pie chart
5. Heat Maps - It will help in showing the trends in matrix
6. Box Plot
7. Tree Maps
8. Geographical / Geographic Maps

Used Case
1. Comparision - Bar or Bubble Graph
2. Distribution - Histogram or Box Plot
3. Composition - Pie-chart, Tree Graph, Bar chart
4. Relationship - scatted plot, network diagram,

The Best Practice for Graph

(i) Choose right chart
(ii) Huge chart
(iii) Keep it simple
(iv) Add context to the graph

Bayesian Linear Regression

→ It is an extension of Linear Regression which apply -ies bayesian inference to estimate the model parameters

→ It provide more interpretable results as the model produce ~~distingui~~ distribution over parameters for traditional linear regression model might give a single predicted values which won't tell the confident ~~abman~~ about the model prediction with the respective results.

→ Bayesian Linear Regression on the contrary. It provide the distribution of possible value reflecting the uncertainty of possible estimating value

→ It is the probablistic prediction which is more useful and effective in making informed decision in risk management lanlysis

→ Ex :- Diease Prediction, Stock Market, Pricing of particular product

Bayesian Approach

→ In this approach, the distribution of possible parameter has been found out.

→ It starts with prior belief about the parameter and update this belief using the data to get posterior distribution